

The velocity potential of a three-dimensional flow field is given by

$$\varphi = A \cos(\Omega t)(2x^2 - y^2 - z^2),$$

where A and Ω are known constants.

1. Determine the velocity field, verifying that $\nabla \wedge \vec{v} = 0$ and $\nabla \cdot \vec{v} = 0$.
2. Compute the streamlines, trajectories and particle paths.
3. Obtain the fluid surface that at the initial instant is a sphere of radius R centered at $(x, y, z) = (-R/2, 0, 0)$.

$$1) \quad v_x = \frac{\partial \varphi}{\partial x} = 4A \cos(\Omega t) x$$

$$v_y = -2A \cos(\Omega t) y$$

$$v_z = -2A \cos(\Omega t) z$$

$$\nabla \wedge \vec{v} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = 2A \cos(\Omega t) \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x & -y & -z \end{vmatrix} = 0$$

$$\nabla \cdot \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = A \cos(\Omega t) (4 - 2 - 2) = 0$$

2) STREAMLINES

$$\frac{dx}{v_x} = \frac{dy}{v_y} = \frac{dz}{v_z} \Rightarrow \frac{dx}{2x} = \frac{dy}{-y} = \frac{dz}{-z} \Rightarrow \begin{matrix} xy^2 = A \\ \frac{y}{z} = B \end{matrix}$$

(e.g. THE STREAMLINE THAT INTERSECTS THE POINT x_0, y_0, z_0 IS GIVEN BY

$$\begin{matrix} xy^2 = x_0 y_0^2 \\ \frac{y}{z} = \frac{y_0}{z_0} \end{matrix}$$

TRAJECTORIES

$$\frac{dx}{dt} = v_x = 4A \cos(\Omega t) x$$

$$\frac{dy}{dt} = v_y = -2A \cos(\Omega t) y$$

$$\frac{dz}{dt} = v_z = -2A \cos(\Omega t) z$$

$$t=0: \begin{matrix} x=x_i \\ y=y_i \\ z=z_i \end{matrix}$$

$$\int \frac{dx}{x} = 4A \int \cos(\Omega t) dt \Rightarrow \frac{x}{x_i} = e^{\frac{4A}{\Omega} \sin(\Omega t)}$$

$$\int \frac{dy}{y} = -2A \int \cos(\Omega t) dt \Rightarrow \frac{y}{y_i} = e^{-\frac{2A}{\Omega} \sin(\Omega t)}$$

$$\int \frac{dz}{z} = -2A \int \cos(\Omega t) dt \Rightarrow \frac{z}{z_i} = e^{-\frac{2A}{\Omega} \sin(\Omega t)}$$

PARTICLE PATHS

ELIMINATING THE TIME

$$e^{\frac{2A}{\Omega} \sin(\Omega t)} = \left(\frac{x}{x_i}\right)^{1/2} = \left(\frac{y}{y_i}\right)^{-1} = \left(\frac{z}{z_i}\right)^{-1}$$

$$xy^2 = x_i y_i^2, \quad \frac{y}{z} = \frac{y_i}{z_i}$$

WHICH ARE ALSO THE STREAMLINES OF THE FLOW!!

3) GIVEN THE TRAJECTORIES $\vec{x} = \vec{x}_T(\vec{x}_i, t)$, THE EQUATION FOR A FLUID SURFACE LOCATED INITIALLY AT $\vec{x} = \vec{x}_i(\alpha, \beta)$ IS CONTAINED IN

INITIAL SURFACE

$$x_i - \frac{R}{2} = R \sin \alpha \cos \beta$$

$$y_i = R \sin \alpha \sin \beta \Rightarrow$$

$$z_i = R \cos \alpha$$

$\vec{x} = \vec{x}_T(\vec{x}_i(\alpha, \beta), t)$. BY ELIMINATING THE PARAMETERS α AND $\beta \rightarrow f(x, y, z, t) = 0$

FLUID SURFACE EQS.

$$\begin{matrix} \frac{x}{R/2 + R \sin \alpha \cos \beta} = e^{\frac{4A}{\Omega} \sin(\Omega t)} \\ \frac{y}{R \sin \alpha \sin \beta} = e^{-\frac{2A}{\Omega} \sin(\Omega t)} \\ \frac{z}{R \cos \alpha} = e^{-\frac{2A}{\Omega} \sin(\Omega t)} \end{matrix}$$

SIMPLIFIED EQ

(ELIMINATING α & β)

$$\left(x e^{-\frac{4A}{\Omega} \sin(\Omega t)} - \frac{R}{2}\right)^2 + \left(y e^{\frac{2A}{\Omega} \sin(\Omega t)}\right)^2 + \left(z e^{\frac{2A}{\Omega} \sin(\Omega t)}\right)^2 = R^2$$